Reflectarray with Tapered Hexagonal Grid Design for Microwave Power Transmission

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Introduction

Space-based wireless power transmission (WPT) requires a transmit antenna that is simultaneously lightweight, foldable for launch, radiation-tolerant, and capable of precise beam steering. These requirements demand a design that maximizes beam efficiency relative to aperture size, while also maintaining consistent sidelobe suppression over mission life.

Typically, a reflectarray achieves beam steering entirely through passive element geometry — each printed copper patch reflects an incoming wave with a phase shift determined by its physical radius. No active electronics are required, becoming an obvious choice for space-based power transmission, as it eliminates the dominant failure mode of active phased arrays in the GEO radiation environment.

This work presents the theoretical design, HFSS unit-cell simulation, and array analysis of a circular-patch element on a hexagonal grid with a passive gaussian aperture taper. It also compares these results to a uniform baseline array, and Taylor/Chebyshev amplitude tapers.

Design Methods

For design parameters, the operating frequency was chosen to be the ISM-band standard of 2.45 GHz. From the frequency, ensuring radiation loss dominates dielectric loss ($Q_{diel} \gg Q_{rad}$), and the dominant mode of a microstrip patch (TM_{11}), calculations for thickness, radius, substrate, spacings, etc. were able to be done.

$$J_1'(k_c\, a_{\text{eff}}) = 0 \implies k_c\, a_{\text{eff}} = \chi_{11}' = 1.8412$$

$$a_{\text{eff}} = a\sqrt{1 + \frac{2h}{\pi a \varepsilon_r}\left[\ln\left(\frac{\pi a}{2h}\right) + 1.7726\right]}$$

Parameter	Value	Theoretical Basis
Substrate	**RO4003C (ε_r = 3.55)**	Q_diel = 370 >> Q_rad
Thickness (h)	**3 mm**	0.01λ ≤ h ≤ 0.05λ
Patch Radius (a)	**18.1 mm**	TM_{11} Resonance & Fringing Correction
Cell Spacing D_nn	**50 mm (0.41λ)**	Grating Lobe Constraint
Element Packing	**Hexagonal**	+15.5% Density vs Square

Unit Cell Simulations

Simulated in Ansys HFSS using an infinite-array Floquet port. For each patch radius from 4–21.5 mm, a full adaptive solve at 2.45 GHz extracts the complex reflection coefficient S_{11}. Near resonance (r ≈ 18.1 mm), the phase transitions steeply through 0°, providing the most phase shift per millimeter of radius change.

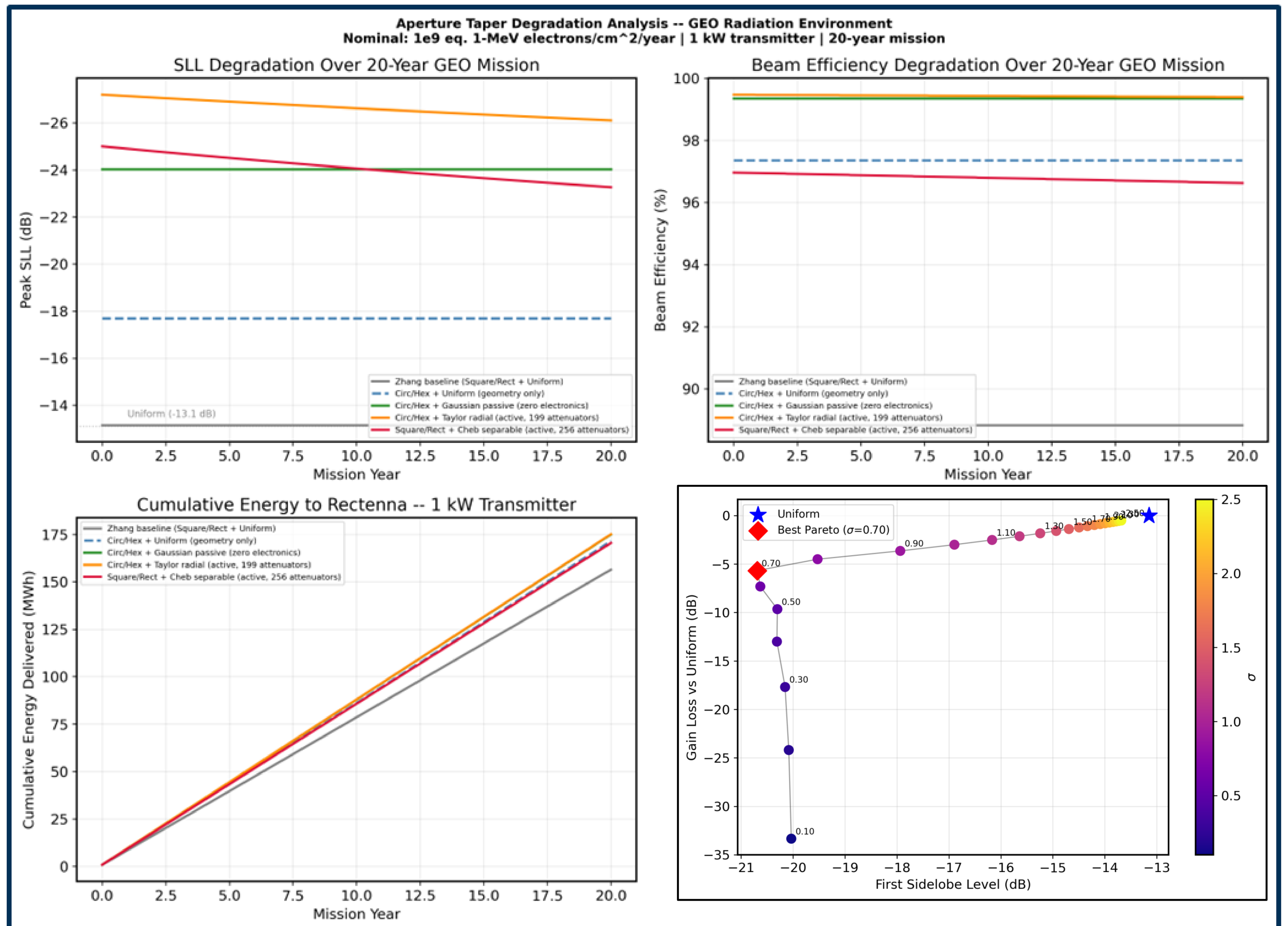
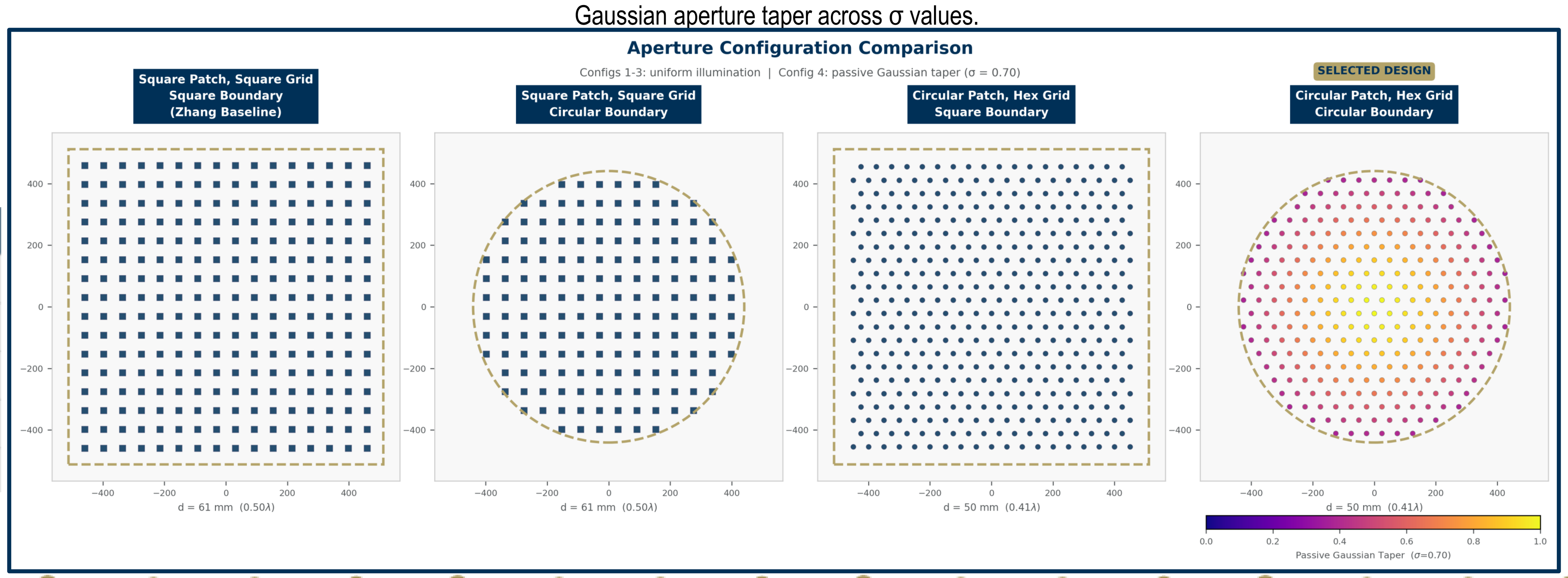


Fig. 1 — 20-year GEO radiation degradation and SLL vs. gain loss tradeoff for Gaussian aperture taper across σ values.



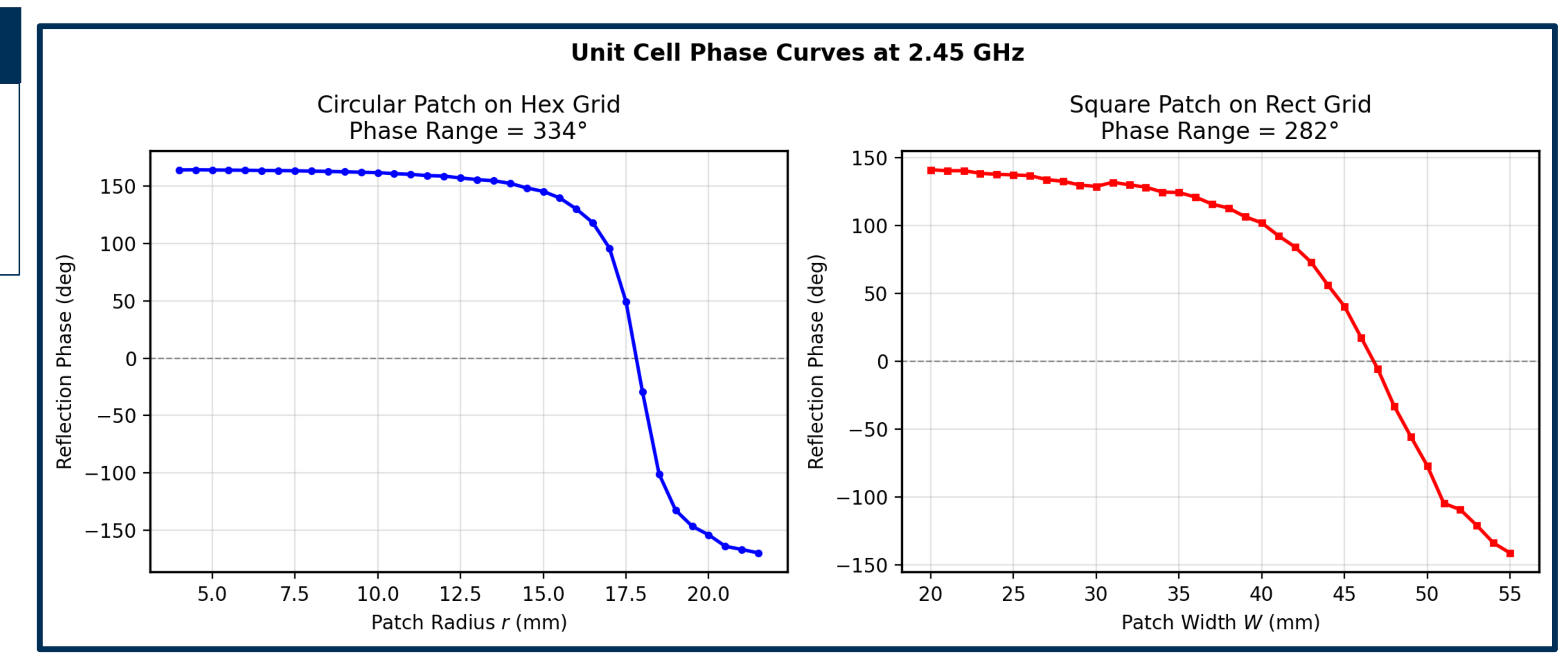


Fig. 2 — Reflection phase vs. patch radius. Circular/hex design achieves 334° range vs. 282° for square baseline (+20%).

Conclusions

The circular patch, hexagonal grid with a passive gaussian aperture taper design proves itself as a capable and robust design choice for a reflectarray. The design achieved a **334° phase range** and **−24 dB sidelobe level** — improvements of +52° and +10.9 dB respectively over the square-patch square-grid baseline of Zhang et al. (2022). The total 334° range exceeds the 300° minimum required for full beam steering coverage across the aperture.

The passive design is radiation-hard, with negligible degradation over a 20-year mission in a GEO radiation environment, while maintaining a high 99.3% beam efficiency.

Future Work

→ Finite array simulation to quantify edge-element mutual coupling
→ Miura-fold crease geometry integration and stowed volume validation
→ Far-field gain measurement with physical feed horn prototype
→ Scaling study: aperture size vs. transmission range tradeoff

Metric	Cir/Hex Design	Zhang Baseline	Improvement
Phase range	**334°**	282°	+52°
Best SLL	**−24.0 dB**	−13.1 dB	+10.9 dB
Beam efficiency	**99.3%**	88.8%	+10.5%
Cross-pol isol.	**> 40 dB**	—	—
Active electronics	**0**	0	—
20-yr energy	**174.9 MWh**	156.4 MWh	+11.8%